



NATIONAL SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2022

MATHEMATICAL LITERACY P2 MARKING GUIDELINE

MARKS: 100

Symbol	Explanation
M	Method
M/A	Method with accuracy
MCA	Method with consistent accuracy
CA	Consistent accuracy
A	Accuracy
C	Conversion
S	Simplification
RT/RG/RM	Reading from a table OR Reading from a graph OR Read from a map
F	Choosing the correct formula
SF	Substitution in a formula
J	Justification
P	Penalty, e.g. for no units, incorrect rounding off etc.
R	Rounding off OR Reason
AO	Answer only
NPR	No penalty for rounding

This marking guideline consists of 8 pages.

QUESTION 1 [20]				
Ques.		Solution	Explanation	Topic & Level
1.1	1.1.1	R40,00 ✓✓ RT	2RT correct value (2)	F L1
	1.1.2	Hours: 8 ✓ C Convert 15 min = $15 \div 60$ = 0,25 + 8 = 8,25 hrs. ✓ A	1C divide by 60 1A adding correct values (2)	M L1
1.2	1.2.1	Departure 14h20 Arrive – 11h30 ✓ MA Lapse time = 2h50 min ✓ A	1MA subtraction time 1A simplified time (2)	M L1
	1.2.2	Cost = R50,00 + (R7,00 x R3,00) ✓ RT = R50,00 + R21,00 R71,00 ✓ A	1RT correct values 1A answer (2)	F L1
1.3	1.3.1	It is the total length of the sides in a shape. ✓✓ OR It is the distance around the outside of the shape. ✓✓ R Accept any other relevant reason.	2R Explanation (2)	M L1
	1.3.2	✓ MA Perimeter rectangular diagram = 210 + 210 + 100 + 100 = 620 cm ✓ CA	1MA adding sides 1CA answer (2)	M L1
	1.3.3	Numerical scale or Ratio scale ✓✓ A	2A concept scale (2)	M&P L1
	1.3.4	Every 1 unit on the map represents 100 units in reality. ✓✓ A	2A Explanation (2)	M&P L1
1.4	1.4.1	R44 ✓✓ A	2A correct road (2)	M&P L1
	1.4.2	Provincial roads serve as feeders to the national roads. ✓✓ A OR Provincial roads also serve as trunk roads in areas where there is no national roads. ✓✓ A Accept any relevant explanation.	2A Explanation (2)	M&P L1
			[20]	

QUESTION 2 [30]				
Ques.		Solution	Explanation	Topic & Level
2.1	2.1.1	Bar scale ✓✓ 2A	2A correct scale (2)	M&P L1
	2.1.2	Width of bar (measured) = 2,1 cm ✓ A Length of animal = 15,1 cm ✓ A $2,1 \text{ cm} = 1,9 \text{ m}$ ✓ M $\frac{15,1}{2,1} \times 1,9 \text{ m} = 13,6619 \text{ m} \checkmark \text{ CA}$ $= 13,7 \text{ m} \checkmark \text{ R}$ OR ✓ A $2,1 \text{ cm} = 1,9 \text{ m}$ ✓ A ✓ M $15,1 \text{ cm} = \frac{15,1 \text{ cm} \times 1,9 \text{ m}}{2,1 \text{ cm}}$ $= 13,6619 \text{ m} \checkmark \text{ CA}$ $= 13,7 \text{ m} \checkmark \text{ R}$	1A measured value for bar width 2,1 cm Accept 2 – 2,3 cm 1A measured value diagram for 15,1 cm 1M multiply by 1,9 m 1CA correct value 1R one decimal number 1A measured value for bar width 2,1 cm 1A measured value diagram for 15,1 cm 1M multiply by 1,9 m 1CA correct value 1R one decimal number Mark according to your measured length. (5)	M&P L2
	2.1.3	1 ton = 1 000 kilogram 20 ton = 20 × 1 000 ✓ A = 20 000 kilogram ✓ A	1A × 1 000 1A correct value (2)	M L1
2.2		48 000 – 30 000 = 18 000 ✓ MA ✓ A Percentage illegal fishing = $\frac{18\,000}{30\,000} \times 100$ $= 60\% \checkmark \text{ A}$	1MA Subtracting correct number of tuna 1A × 100 1A correct percentage (3)	F L3

2.3	2.3.1	M9 ✓ RT M18 ✓ RT	2RT correct roads (2)	M&P L1
	2.3.2	N2 ✓✓ RT	2 RT (2)	M&P L1
	2.3.3	R75 ✓✓ RT	2 RT (2)	M&P L1
	2.3.4	6 provincial roads ✓✓ 2A OR R75; R102; R334; R335; R367; R368 ✓✓ 2A	2A correct number 2A correct list of roads (2)	M&P L2
	2.3.5	Measured distance : 5 cm ✓ A 5 cm : 13 km ✓ M 5 cm : 1300 000 cm ✓ C $\frac{5 \text{ cm}}{5 \text{ cm}} : \frac{1 \text{ 300 000 cm}}{5 \text{ cm}} \checkmark \text{ S}$ 1 : 260 000 Yes, his statement is valid. ✓✓ 2J	1A measured value 1M ratio concept 1C conversion 1S simplify values 2J conclusion (6)	M&P L4
	2.3.6	Time = $\frac{13 \text{ km}}{80 \text{ km /h}} \checkmark \text{ SF}$ = 0,1625 hrs × 60 = 9,75 min ✓ C = 0,75 × 60 = 45 seconds ✓ C Total time = 9 min 45 seconds ✓ S	1SF correct substitution 1C converting hrs to min 1C converting min to seconds 1S simplifying total time (4)	M L2
			[30]	

QUESTION 3 [23]				
Ques.		Solution	Explanation	Topic & Level
3.1	3.1.1	Perimeter = 2 (length + breadth) ✓SF = 2 (8 500 mm + 3 500 mm) = 2 (12 000 mm) ✓ = 24 000 mm ✓	1SF correct values 1S simplification 1CA answer (3)	M L2
	3.1.2	Area of floor = Length × Width ✓ C = 8,5 m × 3,5 m ✓ = 29,75 m ² ✓	1C convert to m 1M multiply 5,8 m with 3,5 m 1CA correct value (3)	M V2
	3.1.3	Volume = length × breadth × height (thickness) = 8,5 m × 3,5 m × 0,1 m ✓ = 2,975 m ³ ✓	2A Reason 1SF substitute correct values 1CA answer for 2,975 m ³ NPR (2)	M L3
	3.1.4	Facing west ✓ and east. ✓	2RT directions (2)	M&P L3

3.2	3.2.1	$\checkmark C$ Area of southern wall = $8,5 \text{ m} \times 5 \text{ m} \checkmark \text{ SF}$ $= 42,5 \text{ m}^2 \checkmark \text{ CA}$ Dimensions of 1 brick (with mortar) Length = $222 + 10 \checkmark \text{ M}$ $= 232 \text{ mm}$ Height = $73 + 10$ $= 83 \text{ mm}$ Area of 1 brick = length $\times$ height $= 232 \text{ mm} \times 83 \text{ mm} \checkmark \text{ SF}$ $= 19\,256 \text{ mm}^2 \div 1\,000\,000 \checkmark C$ $= 0,019\,256 \text{ m}^2 \checkmark \text{ CA}$ No. of bricks = $\frac{\text{Area of the wall in m}^2}{\text{Area of one brick in m}^2}$ $= \frac{42,5 \text{ m}^2}{0,019256 \text{ m}^2} \checkmark \text{ SF}$ $= 2\,207,10$ $= 2\,208 \text{ bricks} \checkmark \text{ CA}$	1C Convert cm to m 1SF correct values 1CA correct value for $42,5 \text{ m}^2$ 1M adding 10 mm 1SF correct values 1C divide by conversion factor 1CA for $0,019256 \text{ m}^2$ 1SF correct values 1CA number of bricks (9)	M L3
	3.2.2	$10\% \times 2\,208 \text{ bricks} = 220,8 + 2\,208 \checkmark \text{ M}$ Total amount of bricks $\approx 2\,428,8$ $\approx 2\,429 \text{ bricks} \checkmark \text{ CA}$ OR $110\% \times 2\,208 = 2\,428,8 \checkmark \text{ S}$ $\approx 2\,429 \text{ number of bricks} \checkmark \text{ CA}$	CA from 3.2.1 1M adding correct no of bricks 1CA total no of bricks 1S simplifying correct number of bricks 1CA total number of bricks (2)	M L2
	3.2.3	2 429 total number of bricks (answer to QUESTION 3.2.2) R4,75 price of one brick $2\,429 \times \text{R}4,75 \checkmark \text{ M}$ $= \text{R}11\,537,75 \checkmark \text{ CA}$	1CA total number of bricks 1M price of one brick 1CA total amount (2)	M L4
			[23]	

QUESTION 4 [27]				
Ques.		Solution	Explanation	Topic & Level
4.1	4.1.1	Scale 1 : 30 $\checkmark$ C $A = \frac{45 \times 1}{30} = 1,5 \text{ cm}$ $\checkmark$ CA $\checkmark$ M $\checkmark$ M $B = \frac{105 \times 1}{30} = 3,5 \text{ cm}$ $\checkmark$ CA	1C conversion to cm 1M divide by 30 1CA value of 1,5 cm 1M using 105 cm 1CA value of 3,5 cm (5)	M&P L4
	4.1.2	Volume = $l \times b \times h$ $\checkmark$ C $= 105 \text{ cm} \times 45 \text{ cm} \times 30 \text{ cm}$ $\checkmark$ SF $= 141\,750 \text{ cm}^3$ $\checkmark$ CA	1C conversion 1SF substitute correct values 1CA Simplification (3)	M L2
	4.1.3	$1 \text{ cm}^3 = 1 \text{ ml}$ $141\,750 \text{ cm}^3 = 141\,750 \text{ ml}$ $\checkmark$ A 1 litre = 1 000 ml $\checkmark$ M $\frac{141\,750 \text{ ml}}{1\,000 \text{ ml}} = 141,75 \text{ litres}$ $0,9 \times 141,75 = 127,575 \text{ litres}$ $\checkmark$ CA	1A ratio concept 1M divide by 1 000 1CA value of 127,575 litres (3)	M L2
4.2	4.2.1	Actual length = $50,4 \text{ mm} \times 150$ $\checkmark$ M $= 7\,560 \text{ mm} \div 1\,000$ $\checkmark$ M $= 7,56 \text{ m}$ $\checkmark$ CA Actual width = $25,6 \text{ mm} \times 150$ $\checkmark$ M $= 3\,840 \text{ mm} \div 1\,000$ $\checkmark$ M $= 3,84 \text{ m}$ $\checkmark$ CA	1M $\times 150$ 1M $\div 1\,000$ 1CA correct value 1M $\times 150$ 1M $\div 1\,000$ 1CA correct value (6)	M L2
	4.2.2	Area of rectangle = length $\times$ breadth $= 7,56 \text{ m} \times 3,84 \text{ m}$ $\checkmark\checkmark$ SF M $= 29,03 \text{ m}^2$ $\checkmark$ CA	1SF correct values 1M multiply values 1CA answer (3)	M L3

	4.3	Probability not winning: $\frac{10\ 000}{10\ 000} - \frac{1000}{10\ 000}$ $\checkmark A$ $= \frac{9\ 000}{10\ 000} \times 100$ $\checkmark A$ $= 90\% \checkmark A$	1A numerator 1A denominator 1A for 90% (3)	P L2
	4.3.1	Unlikely $\checkmark \checkmark A$	2A Explanation (2)	P L1
	4.3.2	10% : 90 % $\checkmark A$ 1 : 9 $\checkmark A$	1A ratio concept 1A simplified values (2)	P L1
			[27]	
			TOTAL: 100	